A Practical Guide to Graph neural network

TOPIC -- GRAPH NEURAL NETWORK

$$a^{(l)} = \sigma! \left(W^{(l)} a^{(l-1)} + b^{(l)}\right)$$

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Water distribution systems can be modelled as graphs, being then a straightforward application of GNN. This kind of algorithm has been applied to water demand forecasting, interconnecting District Metered Areas(DMAs) to improve the forecasting capacity. Other application of this algorithm on water distribution modelling is the development of metamodels.

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where p {\displaystyle \mathbf {p} } is a learnable projection vector. The projection vector p {\displaystyle \mathbf {p} } computes a scalar projection value for each graph node. The top-k pooling layer can then be formalised as follows:

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where φ {\displaystyle \phi } and ψ {\displaystyle \psi } are differentiable functions (e.g., artificial neural networks), and ■ {\displaystyle \bigoplus } is a permutation invariant aggregation operator that can accept an arbitrary number of inputs (e.g., element-wise sum, mean, or max). In particular, φ {\displaystyle \phi } and ψ {\displaystyle \psi } are referred to as update and message functions, respectively. Intuitively, in an MPNN computational block, graph nodes update their representations by aggregating the messages received from their neighbours. The outputs of one or more MPNN layers are node representations h u {\displaystyle \mathbf {h} _{u}} for each node u ∈ V {\displaystyle u\in V} in the graph. Node representations can be employed for any downstream task, such as node/graph classification or edge prediction. Graph nodes in an MPNN update their representation aggregating information from their immediate neighbours. As such, stacking n {\displaystyle n} MPNN layers means that one node will be able to communicate with nodes that are at most n {\displaystyle n} "hops" away. In principle, to ensure that every node receives information from every other node, one would need to stack a number of MPNN layers equal to the graph diameter. However, stacking many MPNN layers may cause issues such as oversmoothing and oversquashing. Oversmoothing refers to the issue of node representations becoming indistinguishable. Oversquashing refers to the bottleneck that is created by squeezing long-range dependencies into fixed-size representations. Countermeasures such as skip connections (as in residual neural networks), gated update rules and jumping knowledge can mitigate oversmoothing. Modifying the final layer to be a fully-adjacent layer, i.e., by considering the graph as a complete graph, can mitigate oversquashing in problems where long-range dependencies are required. Other "flavours" of MPNN have been developed in the literature, such as graph convolutional networks and graph attention networks, whose definitions can be expressed in terms of the MPNN formalism.

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h u = ■ k = 1 K σ (∑ v ∈ N u α u v k W k x v) {\displaystyle \mathbf {h} _{u}={\overset {K}{\underset {k=1}{\Big \Vert }}}\sigma \left(\sum _{v\in N_{u}}\alpha _{uv}^{k}\mathbf {W} ^{k}\mathbf {x} _{v}\right)}

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To represent an image as a graph structure, the image is first divided into multiple patches, each of which is treated as a node in the graph. Edges are then formed by connecting each node to its nearest neighbors based on spatial or feature similarity. This graph-based representation enables the application of graph learning models to visual tasks. The relational structure helps to enhance feature extraction and improve performance on image understanding.

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Social networks are a major application domain for GNNs due to their natural representation as social graphs. GNNs are used to develop recommender systems based on both social relations and item relations.

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Gated graph sequence neural network The gated graph sequence neural network (GGS-NN) was introduced by Yujia Li et al. in 2015. The GGS-NN extends the GNN formulation by Scarselli et al. to output sequences. The message passing framework is implemented as an update rule to a gated recurrent unit (GRU) cell. A GGS-NN can be expressed as follows:

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Message passing layers are permutation-equivariant layers mapping a graph into an updated representation of the same graph. Formally, they can be expressed as message passing neural networks (MPNNs). Let G = (V , E) {\displaystyle G=(V,E)} be a graph, where V {\displaystyle V} is the node set and E {\displaystyle E} is the edge set. Let N u {\displaystyle N_{u}} be the neighbourhood of some node u ∈ V {\displaystyle u\in V} . Additionally, let x u {\displaystyle \mathbf {x} _{u}} be the features of node u ∈ V {\displaystyle u\in V} , and e u v {\displaystyle \mathbf {e} _{uv}} be the features of edge (u , v) ∈ E {\displaystyle (u,v)\in E} . An MPNN layer can be expressed as follows:

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GNNs have become a primary tool for modeling atomic interactions in computational chemistry and materials science, where molecules and crystal structures are naturally represented as graphs with atoms as nodes and chemical bonds or spatial proximity as edges. A key advantage over traditional quantum chemistry methods such as density functional theory (DFT) is the reduction in computational cost from $O(n^3)$ to $O(n)$, enabling simulations at previously impractical scales. Early GNN architectures for this domain include SchNet (2017), which introduced continuous-filter convolutional layers for learning on 3D molecular structures, and DimeNet (2020), which incorporated angular information between atoms to improve geometric expressivity. Subsequent equivariant architectures such as NequIP and MACE, which respect the rotational and translational symmetries of physical systems, achieved significant accuracy improvements on standard benchmarks including QM9, MD17, and the Materials Project. Large open datasets have driven rapid progress in this area. The Open Catalyst 2020 (OC20) dataset, comprising over 260 million DFT calculations of adsorbate-catalyst surface interactions, established a widely used benchmark for energy and force prediction tasks relevant to catalyst discovery.. The Materials Project and Alexandria databases have similarly provided large-scale crystal structure data for training universal interatomic potentials. More recent datasets including OMat24 and OMol25 have extended coverage to organic molecules and complex materials at greater computational scale. These datasets have enabled the development of universal [[machine-learned interatomic potential|machine learning interatomic potentials]] (MLIPs) — models trained across diverse chemical domains rather than on system-specific data. Examples include CHGNet, M3GNet, MACE-MP, and Meta FAIR's Universal Models for Atoms (UMA), the latter trained on approximately 500 million atomic structures spanning molecules, materials, and catalysts. GNoME, developed by Google DeepMind, applied GNNs to predict the stability of crystal structures, reporting the identification of millions of new stable candidate materials.